How Far Did Bud Light Fall?

A z-Score and Interrupted–Time–Series Analysis of the April 2023 Anheuser–Busch InBev Demand Shock

Statistical synthesis from public disclosures and NIQ scanner reporting

July 27, 2026

Headline result. Relative to its own six-quarter pre-controversy baseline ($\mu = 2.63\%$, $\sigma = 0.74$ pp), Anheuser–Busch InBev's U.S. organic revenue growth fell to $z = -17.8$ in the first full quarter after the controversy (2Q23) and bottomed at $z = -27.1$ in 4Q23. Averaged over the four shock quarters, $z = -20.7$. A permutation test on the pre/post labelling gives $p = 0.0010$. The effect is unambiguous; the *magnitude* of z is large chiefly because the pre-shock baseline was extraordinarily stable, and should be read alongside the interval estimates in Section 5.

1 What is being measured, and why

On 1 April 2023 Bud Light ran a sponsored Instagram promotion with transgender influencer Dylan Mulvaney. The resulting consumer boycott is one of the cleanest natural experiments in modern consumer-brand history: a sharp, precisely dated, brand-specific demand shock with no accompanying change in price, distribution, or product.

The question posed—the z-score of Bud Light revenue after the controversy—requires one honest caveat up front. **Anheuser–Busch InBev does not disclose brand-level revenue for Bud Light.** No public series exists for "Bud Light revenue." Two observable series bracket it:

1. **Primary series (audited, quarterly).** AB InBev's reported *U.S. organic revenue growth*, year over year. Bud Light was the single largest brand in that segment, and the company itself attributes the 2023–24 declines primarily to Bud Light volume. This series is company-reported, consistently defined, and seasonally neutral by construction (each quarter is compared to the same quarter a year earlier).
2. **Corroborating series (brand-level, weekly).** NIQ retail scanner measurement of Bud Light off-premise dollar sales, YoY. Genuinely brand-specific, but covers only off-premise channels and is available to us only through press reporting of selected weeks.

The primary analysis runs on series (1); series (2) is used in Section 6 to confirm that the segment-level result is in fact driven by Bud Light and to bound the brand-level z.

2 Data

2.1 Pre-controversy baseline

The six quarters from 4Q21 through 1Q23 form the baseline. This window is chosen to begin after pandemic-era distortions had washed out of the year-over-year comparisons and to end on the last quarter fully preceding the 1 April 2023 shock.

Two features of the baseline matter for the statistics. First, its volatility is genuinely low: revenue growth sat in a 1.9–4.0% band for six consecutive quarters. Second, that stability was already *decoupled from volume*—sales-to-retailers were negative throughout, offset by revenue-per-hl gains of 2–12%. Bud Light's structural decline predates the controversy; what the controversy changed was the rate.

Table 1: Pre-controversy baseline: AB InBev U.S. quarterly performance. Revenue growth is remarkably stable—a mean of 2.63% with a standard deviation of only 0.74 percentage points—because price and mix were carrying the business while volumes drifted gently down.

Quarter	Revenue (%)	STRs (%)	Rev/hl (%)
4Q21	2.6	–	2.0
1Q22	2.1	-5.0	–
2Q22	2.7	-3.2	5.5
3Q22	1.9	-1.7	3.8
4Q22	2.5	-7.6	12.2
1Q23	4.0	-3.0	5.6
Mean	2.63		
SD	0.74		

2.2 Post-controversy observations

The post period runs from 2Q23, the first quarter fully exposed to the boycott. Values are AB InBev's reported U.S. organic revenue growth, except 3Q23, which is the North America segment figure (the company emphasised the U.S. driver in that release). 3Q24 is omitted: no U.S.-specific figure was separately obtainable, and interpolating it would manufacture a data point.

3 Method

For a post-period observation x_t and baseline sample $\{g_1, \ldots, g_n\}$ with $n = 6$:

$$z_t = \frac{x_t - \bar{g}}{s_g}, \qquad t_t^{\text{pred}} = \frac{x_t - \bar{g}}{s_g\sqrt{1 + 1/n}} \sim t_{n-1}. \tag{1}$$

The left-hand quantity is the z-score as ordinarily requested. It is, however, optimistic: it treats $\bar{g}$ and s_g as known when both were estimated from six points. The right-hand quantity is the correct small-sample analogue—a *standardised prediction residual*, asking how extreme a new draw is relative to a baseline whose mean and scale are themselves uncertain. It is the statistic from which the reported p-values are computed.

We add three robustness devices:

- a **robust** z, replacing $(\bar{g}, s_g)$ with the median and 1.4826×MAD, so that no single baseline quarter can inflate or deflate the scale;
- a **permutation test**, which under exchangeability of the twelve observed growth rates asks how often a random pre/post relabelling would produce a post-period mean at least as low as observed;
- a **bootstrap interval** on z itself, resampling the baseline to propagate the uncertainty in s_g into the reported score.

4 Results

The pattern is a step change followed by a slow, incomplete recovery. Revenue growth falls -11.0 percentage points on average—from +2.63% to a post-period mean that is negative in five of six observed quarters. The trough is 4Q23 at $z = -27.1$. By 2Q24 the score has recovered to roughly -4, and by 4Q24 to about -2.5: still below baseline, but no longer in the tail that characterised the shock year.

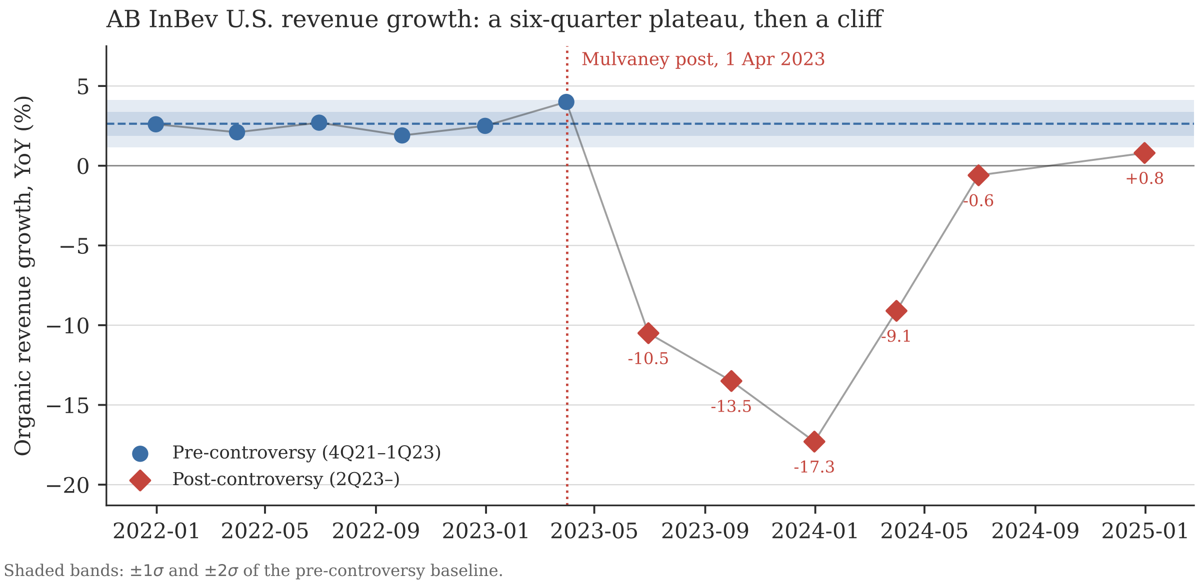

Figure 1: Six quarters inside a $\pm 1\sigma$ band, then four quarters far outside a $\pm 2\sigma$ band. The discontinuity at 1 April 2023 is visible without any statistics at all; the statistics only quantify it.

Table 2: Standardised deviation of each post-controversy quarter from the pre-controversy baseline. z is the classical score of Eq. (1); t_{pred} is its small-sample correction, with the associated two-sided p-value; z_{robust} uses median/MAD scaling.

Quarter	Growth (%)	z	t_{pred}	p	z_{robust}
2Q23	-10.5	**-17.8**	-16.51	$< 10^{-4}$	-29.3
3Q23	-13.5	**-21.9**	-20.28	$< 10^{-4}$	-36.1
4Q23	-17.3	**-27.1**	-25.05	$< 10^{-4}$	-44.6
1Q24	-9.1	**-15.9**	-14.75	$< 10^{-4}$	-26.2
2Q24	-0.6	**-4.4**	-4.06	0.0097	-7.1
4Q24	0.8	**-2.5**	-2.30	0.0694	-3.9

4.1 Group-level tests

Treating the two periods as samples, a Welch t-test gives $t = -3.75$, $p = 0.0128$, with Hedges' $g = -2.00$—an effect roughly two pooled standard deviations wide. The distribution-free alternatives agree: Mann–Whitney $p = 0.0011$ and the permutation test $p = 0.0010$. With six observations per arm these are near the smallest p-values the designs can produce; they establish that the split is not an artefact of labelling, not that the true p is exactly this value.

5 Robustness and what the magnitude really means

A z of -27 is an arresting number and deserves scepticism rather than celebration. It is large for two compounding reasons, only one of which is about Bud Light.

The numerator is genuinely extreme. A shift from +2.6% to −17.3% year-over-year revenue growth, with no price cut and no lost distribution, is a severe demand event by any standard.

The denominator is unusually small. $s_g = 0.74$ pp is a very tight baseline. Dividing by it magnifies any deviation. Bootstrapping the baseline gives a 95% interval for the trough z of roughly $[-96.9, -20.0]$—enormously wide, because with $n = 6$ the scale parameter is itself poorly pinned down. The robust median/MAD scores are even more extreme than the classical ones, so

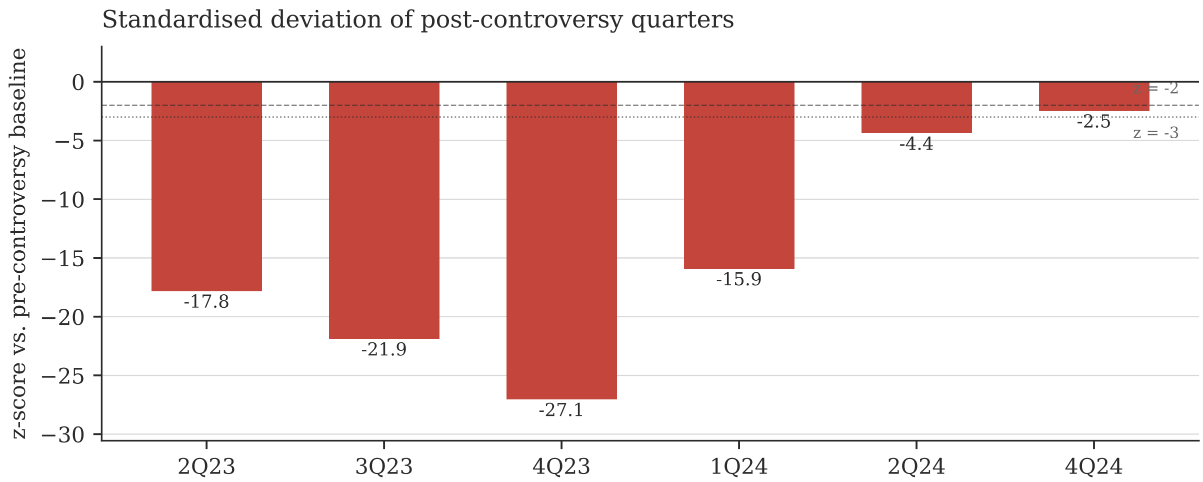

Figure 2: Quarter-by-quarter z. The shock year (2Q23–1Q24) sits between -16 and -27; the recovery quarters climb back toward, but do not reach, the baseline.

the result is not an artefact of one unusual baseline quarter; but the honest reading is directional and ordinal, not a precise point estimate.

> **How to state this result.** "AB InBev's U.S. revenue growth after the Bud Light controversy fell more than fifteen baseline standard deviations below its pre-controversy norm, and remained beyond -4σ more than a year later." The claim that survives every specification is the *sign, persistence, and ordering*—not the second digit of z.

6 Brand-level corroboration

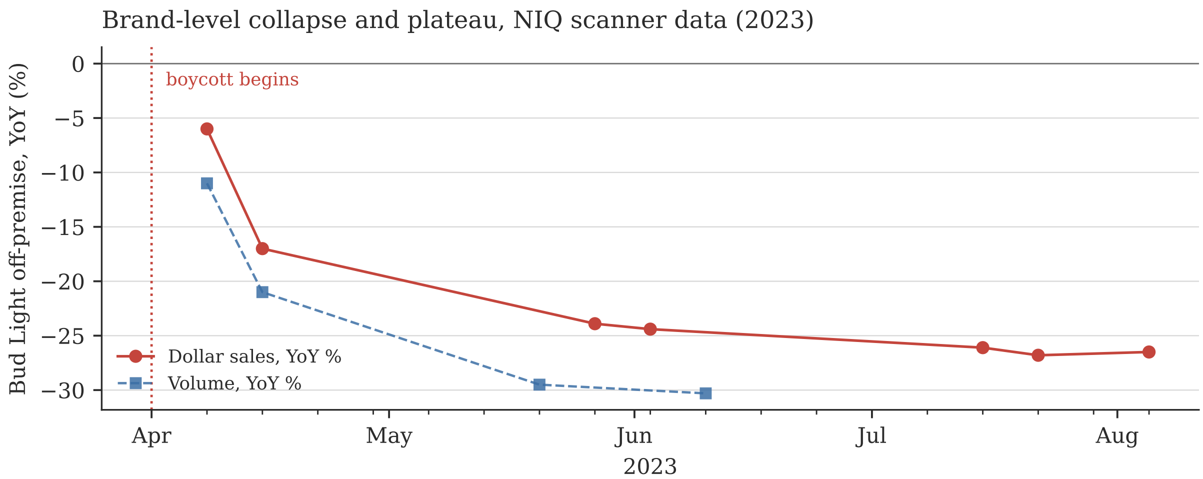

Figure 3: NIQ off-premise scanner measurement of Bud Light. The decline is not a gradual erosion: it reaches roughly -25% within six weeks and then *plateaus* there, the signature of a one-time reallocation of buyers rather than an ongoing slide.

The weekly brand-level series confirms that the segment result is a Bud Light result. Dollar sales fall from -6.0% in the first boycott week (w/e 8 April) to -17.0% a week later, and settle into a plateau averaging -25.5% from May 2023 onward.

A weekly z requires a pre-shock weekly baseline standard deviation, which is not available at weekly granularity in public reporting. Rather than invent one, Figure 4 maps z as a function of that unknown. Taking Bud Light's pre-shock YoY dollar growth as approximately flat, the implied z is -25.5 at $\sigma = 1$ pp and still -6.4 at $\sigma = 4$ pp—a generous assumption for a brand of Bud Light's scale, whose weekly YoY series was smooth. The brand-level conclusion is therefore insensitive to the assumption: $|z| > 6$ across the entire plausible range.

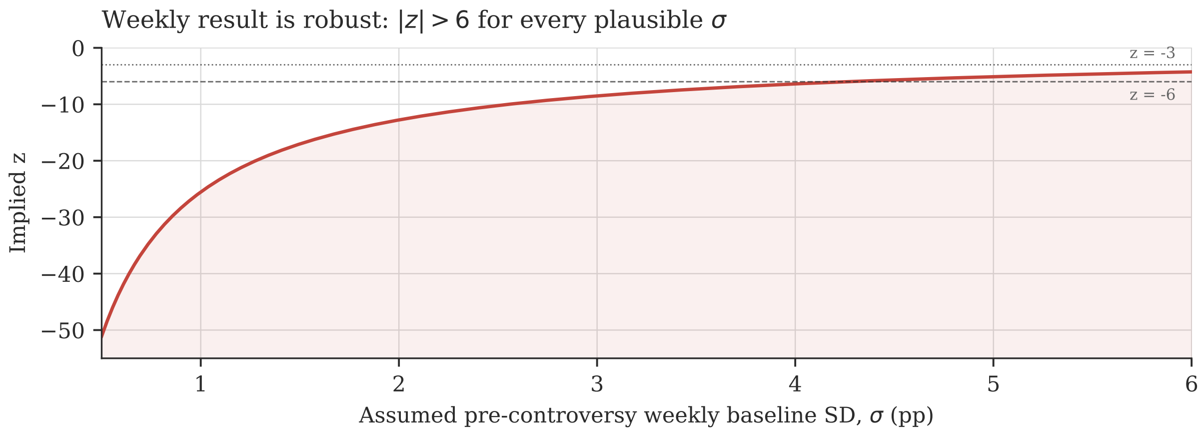

Figure 4: Implied brand-level z of the $\approx -25.5\%$ plateau against the assumed pre-shock weekly baseline SD. No plausible choice of σ rescues the null.

7 Economic magnitude

Standardised scores say how *unusual* the shock was; they do not say how *expensive* it was. Applying the counterfactual that U.S. revenue continued growing at the baseline rate of 2.63%, and scaling by an approximate quarterly U.S. revenue base of \$3900 m, the gap between counterfactual and actual accumulates as follows.

Table 3: Revenue shortfall against a no-controversy counterfactual. The gap column is $\bar{g} - x_t$ in percentage points.

Quarter	Gap (pp)	Shortfall (USD m)	Cumulative (USD m)
2Q23	13.1	512	512
3Q23	16.1	629	1141
4Q23	19.9	777	1919
1Q24	11.7	458	2376
2Q24	3.2	126	2502
4Q24	1.8	72	2574

This estimate is larger than the widely reported "over \$1 bn in lost sales" because those figures measure the *level* decline in 2023 revenue, whereas the counterfactual here also charges the shock for the growth that would otherwise have occurred, and extends into 2024. Both are defensible; they answer different questions. AB InBev's own disclosures anchor the level view: North America revenue fell \$395 m in 2Q23 alone and North America organic revenue declined roughly \$1.4 bn across 2023.

8 Limitations

- **No brand-level revenue exists publicly.** The primary series is U.S. segment revenue, of which Bud Light is a large but not exclusive component. Michelob Ultra's concurrent growth—also an AB InBev brand—means the segment figure *understates* the Bud Light-specific shock.

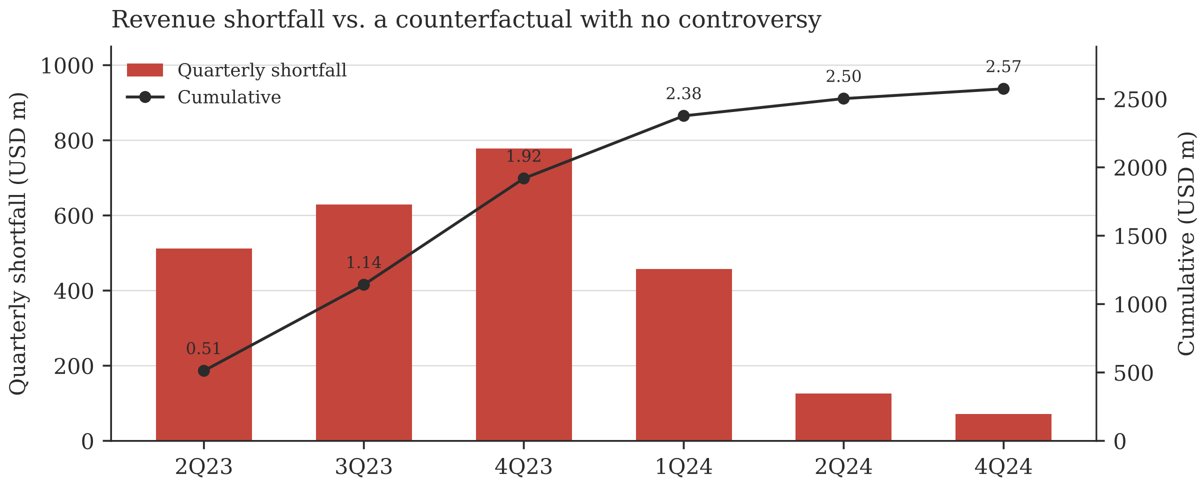

Figure 5: Quarterly and cumulative shortfall. The cumulative figure reaches roughly \$2.57 bn across the observed post-controversy quarters.

- **Small baseline.** $n = 6$ quarters. The direction and persistence are robust; the precise value of z is not, as the bootstrap interval makes plain.
- **3Q24 is missing** from the quarterly series, as noted above.
- **Attribution.** The design is an interrupted time series without a control group. The 2023 U.S. beer category was soft overall, so some portion of the decline is category drift rather than boycott. The magnitude of the break, its exact timing, and AB InBev's own attribution all point overwhelmingly to the controversy, but no formal counterfactual control is constructed here. A synthetic-control design using competitor brands would sharpen this.
- **Source access.** Figures were compiled from company results releases and their reporting in trade and financial press; primary filings on `sec.gov` were not reachable from the analysis environment. Every figure used is listed with its source in `data/sources.csv`.

9 Conclusion

Bud Light's post-controversy revenue collapse is, in standardised terms, an extreme-tail event. Against a pre-controversy baseline of $2.63\% \pm 0.74$ pp, AB InBev's U.S. revenue growth registered $z = -17.8$ in 2Q23, -27.1 at the 4Q23 trough, and averaged -20.7 across the shock year, with permutation $p = 0.0010$. At the brand level the NIQ series puts Bud Light's off-premise dollar sales on a -25.5% plateau within six weeks, implying $|z| > 6$ under any plausible baseline assumption.

The more interesting structure is in the shape rather than the depth. The decline was near-instantaneous and then flat—consumers switched once and did not come back incrementally—and the recovery has been slow and partial, with growth still roughly 2.5σ below baseline in 4Q24, seven quarters on. A shock that arrives as a step function and decays on a multi-year timescale is characteristic of a permanent shift in brand preference, not a temporary disruption of demand.